

THE INSTANTON HOMOLOGY OF THE $(-2, 3, q)$ PRETZEL KNOTS AND COMPUTED BOUNDING COCHAINS IN THE PILLOWCASE

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ABSTRACT. We prove that the reduced singular instanton knot homology of the pretzel knots $P(-2, 3, q)$ has rank $q + 2$ for every odd $q \geq 3$: the Alexander polynomials of the family, computed in closed form by a skein recursion (they are Hironaka’s Lehmer-like polynomials, with all coefficients in $\{0, \pm 1\}$), give the lower bound $q + 2$, and Manion’s closed-form reduced Khovanov homology gives the matching upper bound. We then compare this against the pillowcase (symplectic) side of the knot Atiyah–Floer program. Working in the immersed-curve combinatorial model of Herald–Kirk and Smith, we reconstruct the pillowcase Lagrangians of the natural tangle decomposition of the family and compute the naive Lagrangian–Floer homology of its members through $q = 13$. The outcome is a sharp experimental law: the naive rank differs from $\text{rank } I^{\natural}$ by *exactly one differential*,

$$\text{naive rank} - \text{rank } I^{\natural} = 2 \operatorname{sgn}(\det K - 3),$$

vanishing for the torus member $q = 3$ and changing direction as the determinant $\det = |q - 6|$ crosses 3 (equivalently, as binary-dihedral traceless characters appear). Finally we compute the bounding cochains — the correction terms conjectured by Cazassus–Herald–Kirk–Kotelskiy — that repair the deficiency: for $q = 5$ a unique two-crossing cochain acting through an immersed quadrilateral, for $q = 7$ a unique single crossing acting through a triangle (each *cancelling* a bigon and raising the rank by 2), and for $q = 11$ single-crossing cochains acting in the *opposite* direction (creating a differential and lowering the rank by 2). These are, to our knowledge, the first explicitly computed nonzero bounding cochains in pillowcase Floer theory, and they realize both directions of the conjectured correction within a single family, with a marked rigidity asymmetry: cancellation admits a unique minimal cochain, creation many. We separate throughout what is proved unconditionally, what is computed within the combinatorial model, and what remains conjectural.

1. INTRODUCTION

1.1. Two homologies for one knot. Let $K \subset S^3$ be a knot. On the gauge-theoretic side sits the reduced singular instanton knot homology $I^{\natural}(K)$ of Kronheimer–Mrowka [8, 9], whose graded Euler characteristic is the Alexander polynomial [10, 8] and which receives a spectral sequence from reduced Khovanov homology [9]; hence

$$\ell(K) := \sum_i |a_i(\Delta_K)| \leq \text{rank } I^{\natural}(K) \leq \dim Kh_r(K) =: u(K). \quad (1)$$

On the symplectic side sits the pillowcase program of Hedden–Herald–Kirk [4, 5]. Present $K = (B^3, T_1) \cup_{(S^2, 4)} (B^3, T_2)$ as a union of two tangles. The traceless $\text{SU}(2)$ character variety of the four-punctured sphere is the *pillowcase*

$$P = (\mathbb{R}/2\pi\mathbb{Z})^2 / \iota, \quad \iota(\gamma, \theta) = (-\gamma, -\theta),$$

a two-sphere with four $\mathbb{Z}/2$ orbifold points and symplectic form $d\gamma \wedge d\theta$ on the smooth locus P^* [11, 4]. Each tangle determines an immersed Lagrangian $\mathcal{R}_{\pi_i}(T_i) \looparrowright P^*$, the intersections $\mathcal{R}_{\pi_1}(T_1) \cap \mathcal{R}_{\pi_2}(T_2)$ are the traceless representations of the knot group, and one forms their Lagrangian–Floer homology — an intended Atiyah–Floer counterpart of $I^{\natural}(K)$.

This naive Floer homology is *not* a knot invariant: figure-eight bubbling at the curves' self-intersections spoils invariance under tangle moves. Cazassus, Herald, Kirk and Kotelskiy [2] conjectured the repair: each tangle carries a *bounding cochain* $b \in CF(\mathcal{R}(T), \mathcal{R}(T))$, an $\mathbb{F}_2$ -combination of the self-intersection points satisfying the Maurer–Cartan equation

$$\sum_{k \geq 0} \mu^k(b, \dots, b) = 0 \quad (2)$$

(μ^k counts immersed $(k+1)$ -gons), and the b -deformed Floer homology $HF((\mathcal{R}(T_1), b_1), (\mathcal{R}(T_2), b_2))$ is conjecturally the knot invariant, equal to $I^\natural(K)$ [2, Conj. 1.1]. Before this work, one bounding cochain had been exhibited (on the earring tangle of $T(4, 5)$ [2, §10]), one had been proven nonzero without being computed (the Conway-sum tangle of $P(-2, 3, 5)$, Smith [13, Thm. 5.9]), and none had been computed.

This paper studies both sides for the pretzel family

$$K_q := P(-2, 3, q) = \text{num}(Q_{-1/2} + Q_{1/3} + Q_{1/q}), \quad q \geq 3 \text{ odd}, \quad (3)$$

whose smallest members are the torus knots $K_3 = T(3, 4) = 8_{19}$ and $K_5 = T(3, 5) = 10_{124}$, and which contains the celebrated $K_7 = P(-2, 3, 7)$. We prove an unconditional theorem on the instanton side, measure the pillowcase side against it, and compute the correcting cochains.

1.2. The instanton side: a theorem.

Theorem 1.1. *For every odd $q \geq 3$,*

$$\text{rank } I^\natural(P(-2, 3, q)) = q + 2.$$

The proof occupies Section 2 and is an assembly of three ingredients, of which only the first requires computation. (i) The Alexander polynomials of the family are, up to units and the substitution $t \mapsto -t$, the Lehmer-like polynomials

$$\Delta_{K_q}(t) \doteq t^g + t^{g-1} - \sum_{j=-(g-3)}^{g-3} t^j + t^{-(g-1)} + t^{-g}, \quad g = \frac{q+3}{2}, \quad (4)$$

with all coefficients in $\{0, \pm 1\}$: two ones at each end, one internal zero on each side, and a solid block of $q - 2$ signed ones in the middle, so $\ell(K_q) = 4 + (q - 2) = q + 2$. We derive (4) from the Conway skein relation applied in the q -twist band, which produces a three-term Chebyshev-type recursion solvable in closed form; the polynomials agree with Hironaka's family [7] (for $q = 7$, (4) is precisely Lehmer's polynomial). (ii) Manion's closed-form computation of the reduced Khovanov homology of non-quasi-alternating 3-strand pretzels [12, Thm. 1.1] gives, for $P(-2, 3, q)$, $\dim Kh_r = 2^2 + (3 - 2)(q - 2) = q + 2$. (iii) The squeeze (1) closes the gap. To our knowledge Theorem 1.1 has not appeared in the literature (the case $q = 5$ is Smith's [13]); it is elementary given [12, 7], but it is exactly the input the pillowcase comparison needs, and it is unconditional — no immersed-curve model and no conjecture enters.

1.3. The pillowcase side: a deficiency law. For the decomposition (3) take $T_1 = \widehat{Q_{-1/2}}$ (with the earring added, so its bounding cochain vanishes [13, Lem. 5.6]) and $T_2 = Q_{1/3} + Q_{1/q}$, with $\gcd(q, 3) = 1$ so that the Conway sum is corner-circle free. We reconstructed the two Lagrangians from the quaternionic representation theory — validating the reconstruction against every number in [13] for $q = 5$ (Section 3.2) — and computed the *naive* ($b = 0$) Lagrangian–Floer homology of the pair for the family members through $q = 13$, at two independent perturbations each. The result (Table 1):

Computation 1.2 (Deficiency law). For $q \in \{3, 5, 7, 11, 13\}$, at every tested perturbation,

$$\text{rank } HF((\mathcal{R}^\natural(T_1), 0), (\mathcal{R}_t(T_2), 0)) - \text{rank } I^\natural(K_q) = 2 \operatorname{sgn}(\det K_q - 3), \quad \det K_q = |q - 6|.$$

That is: the naive pillowcase homology is *exact* for the torus member $q = 3$; it is *too small by one differential* for $q = 5, 7$ ($\det = 1$); and it is *too large by one differential* for $q = 11, 13$ ($\det \geq 5$).

q	knot	$\det$	naive rank HF	rank $I^\natural$	naive $-I^\natural$	correction
3	$8_{19} = T(3, 4)$	3	5	5	0	none ($b = 0$)
5	$10_{124} = T(3, 5)$	1	5	7	-2	cancel a bigon
7	$P(-2, 3, 7)$	1	7	9	-2	cancel a bigon
11	$P(-2, 3, 11)$	5	15	13	+2	create a differential
13	$P(-2, 3, 13)$	7	17	15	+2	(predicted)

TABLE 1. The family (3). Naive ranks are computed in the combinatorial model at two independent perturbations each (identical values). $\text{rank } I^\natural = q + 2$ is Theorem 1.1, unconditional. The deficiency is ± 2 or 0 — always exactly one differential — with the sign governed by $\det \geq 3$; equivalently by the number $d = (\det - 1)/2$ of binary-dihedral traceless characters ($d = 0$: deficit; $d = 1$: exact; $d \geq 2$: excess).

We emphasize the register: the naive ranks are computed within the immersed-curve combinatorial model (as is every computation in this subject); the law of Computation 1.2 is verified at five members and two perturbations each, and is a conjecture beyond them. What makes it sharp is Theorem 1.1: without the unconditional value of $\text{rank } I^\natural$ the deficiency could not be measured at all. Note also what the law killed: the naive rank is *not* monotone in any obvious knot complexity (it is 5, 5, 7, 15, 17), and simpler candidate laws (e.g. deficiency proportional to $\det - 3$) are refuted by $q = 13$.

1.4. The bounding cochains: both directions realized. The conjectural repair must therefore *add* rank 2 for $q = 5, 7$ and *remove* rank 2 for $q \geq 11$. Over $\mathbb{F}_2$, adding rank 2 to the homology means *cancelling* one differential (the deformation removes a bigon from the differential), and removing rank 2 means *creating* one. We compute cochains doing both.

Computation 1.3 (The cochains). Within the combinatorial model, with b supported on the self-intersections of $\text{blue} = \mathcal{R}_t(Q_{1/3} + Q_{1/q})$:

- (i) ($q = 5$; *deficit*.) The unique minimal bounding cochain with deformed rank $7 = \text{rank } I^\natural$ is $b = s_A + s_B$, two self-intersections at $(\gamma, \theta) \approx (0.03, 1.27)$ and $(3.06, 4.98)$, one on each pillowcase seam; it acts through a single immersed *quadrilateral* (μ^3) that cancels one of the two bigons.
- (ii) ($q = 7$; *deficit*.) The unique minimal bounding cochain with deformed rank $9 = \text{rank } I^\natural$ is a *single* self-intersection at $\approx (0.05, 5.41)$; it acts through an immersed *triangle* (μ^2) that cancels one of the three bigons.
- (iii) ($q = 11$; *excess*.) There are *fifty-five* single-crossing Maurer–Cartan-valid bounding cochains with deformed rank $13 = \text{rank } I^\natural$; each acts through an immersed triangle that *creates* one new differential, and all but a handful of the created differentials flow into one distinguished generator.

All values are stable under change of the perturbation parameters, and each cochain satisfies the Maurer–Cartan equation (2) to the computed orders (for $q = 5$ the equation is vacuous on small support; for $q = 7, 11$ the self-differential μ^1 is nonzero and the support crossings are μ^1 -closed).

These are the first explicitly computed nonzero bounding cochains in pillowcase Floer theory, and items (i)/(ii) versus (iii) realize the two directions of the conjectured correction within one family. Two structural contrasts deserve emphasis. First, the *polygon order* of the correction is not uniform even within one direction: cancellation is carried by a quadrilateral at $q = 5$ (no triangle reaches a bigon there) but by a triangle at $q = 7$. Second, there is a marked *rigidity asymmetry* between the directions: at the deficit members the minimal cochain is unique — a cancelling polygon must land exactly on an existing bigon — while at the excess member fifty-five distinct minimal cochains work, because creating a differential merely requires an immersed triangle through any suitable crossing. Rank-matching alone thus pins the cochain at deficit members but not at excess members; selecting *the* cochain of the tangle at an excess member requires the (conjectural) naturality data, not just the rank.

1.5. Method and honesty. Nothing on the pillowcase side was taken from [13] as input. Both Lagrangians were reconstructed from first principles — the quaternionic representation theory, Smith’s fiber-product description of the Conway sum and his cut-and-paste resolution of the seam circles, and the Herald–Kirk earring — and validated against the published output of [13] at $q = 5$ (nine generators, two bigons, rank 5, and the worked resolution example) before anything new was computed. The three registers are: (a) *proved unconditionally* — Theorem 1.1; (b) *computed within the model* — the naive ranks, the deficiency law at its five members, the cochains, all by a self-contained program included with the source, with polygon counts by validated winding-number tests; (c) *conjectural* — the faithfulness of the combinatorial model [2, Conj. 1.1] (under which the computed cochains are the values of the invariant-theoretic ones), and the deficiency law beyond $q = 13$. We prove no theorem about the analytic Atiyah–Floer correspondence.

Section 2 proves Theorem 1.1. Section 3 assembles the Lagrangians and records the validation gates and the naive-rank computations. Section 4 states the combinatorial Floer package. Section 5 computes the three cochains. Section 6 discusses the law, the threshold at $\det = 3$, and the open problems.

Acknowledgements. The pillowcase model and the geometry reconstructed here are due to Hedden–Herald–Kirk [4, 5], Cazassus–Herald–Kirk–Kotelskiy [2], Herald–Kirk [6], and Smith [13]; the closed-form Khovanov input is Manion’s [12], and the Alexander polynomials belong to Hironaka’s family [7].

2. THE INSTANTON RANK OF THE FAMILY

2.1. The Alexander polynomials.

Proposition 2.1. *For every odd $q \geq 3$, the Alexander polynomial of $P(-2, 3, q)$ is given, up to multiplication by $\pm t^k$ and the substitution $t \mapsto -t$, by (4). In particular all coefficients lie in $\{0, \pm 1\}$, exactly $q + 2$ of them are nonzero, and*

$$\ell(P(-2, 3, q)) = \sum_i |a_i(\Delta)| = q + 2.$$

Proof. Apply the Conway skein relation $\nabla(L_+) - \nabla(L_-) = z \nabla(L_0)$ at a crossing of the q -twist band of the standard pretzel diagram. Switching the crossing replaces q by $q - 2$; smoothing it replaces q by $q - 1$ (a two-component pretzel link). Writing a_q, b_q for the Conway polynomials of the odd (knot) and even (link) members, the relation gives $a_q = a_{q-2} + sz b_{q-1}$ and $b_{q-1} = b_{q-3} + sz a_{q-2}$ with a fixed orientation sign s ; eliminating b yields the three-term recursion

$$a_{q+2} = (2 + z^2) a_q - a_{q-2} \quad (q \text{ odd}), \quad (5)$$

independent of s . Substituting $z^2 = t - 2 + t^{-1}$, the multiplier is $t + t^{-1}$, so (5) is a Chebyshev-type recursion whose characteristic roots are $t^{\pm 1}$; its solution is determined by the seeds $q = 3, 5$, which are the torus knots $T(3, 4)$, $T(3, 5)$ with classical Alexander polynomials. The normalization ambiguity (each Δ is defined up to $\pm t^k$, and this family satisfies Hironaka's $t \mapsto -t$ twist [7]) is fixed by validation: among the finitely many normalizations of the seeds, exactly one makes (5) reproduce $\Delta_{P(-2, 3, 7)}$ — Lehmer's polynomial, known independently [7] — and it does so exactly, in all eleven coefficients. In that normalization one checks directly that the right side of (4) satisfies (5) (multiplication by $t + t^{-1}$ shifts the two-spike ends and extends the middle block by one on each side; subtracting the previous member restores the pattern), and agrees with the seeds; by induction (4) holds for all odd q . The coefficient count is $2 + 2 + (2(g - 3) + 1) = 2g - 1 = q + 2$. $\square$

Remark 2.2. For $q = 7$, (4) reads $t^5 + t^4 - t^2 - t - 1 - t^{-1} - t^{-2} + t^{-4} + t^{-5}$: Lehmer's polynomial, of Mahler measure ≈ 1.17628 , smallest known. Hironaka [7] proved that the $(-2, 3, q)$ pretzels realize the minimal-Mahler-measure family; Proposition 2.1 is a self-contained derivation of exactly the data we need $(\Sigma|a_i|)$, included so that Theorem 1.1 does not rest on a translation of conventions.

2.2. The Khovanov bound and the squeeze. Manion [12, Thm. 1.1] computes the reduced Khovanov homology of every non-quasi-alternating 3-strand pretzel. For $P(-p, q, r)$ with p even, the total rank is $p^2 + (q - p)(r - p)$; the knots $P(-2, 3, q)$, $q \geq 3$ odd, are in the non-quasi-alternating range ($2 \leq 3 \leq q$), so

$$u(P(-2, 3, q)) = \dim Kh_r(P(-2, 3, q)) = 4 + (3 - 2)(q - 2) = q + 2.$$

(For the members with known Khovanov homology this agrees: 8_{19} has $\dim Kh_r = 5$ and 10_{124} has 7.)

Proof of Theorem 1.1. The graded Euler characteristic of $I^\natural$ is the Alexander polynomial [8, 10], so $\text{rank } I^\natural \geq \ell = q + 2$ by Proposition 2.1; the Kronheimer–Mrowka spectral sequence from reduced Khovanov homology [9] gives $\text{rank } I^\natural \leq u = q + 2$. Hence $\text{rank } I^\natural = q + 2$. $\square$

Remark 2.3. The same squeeze is how Smith identifies $\text{rank } I^\natural(P(-2, 3, 5)) = 7$ [13, proof of Thm. 5.9]; the content added here is that Hironaka's and Manion's closed forms make it uniform in q . We also note $\det P(-2, 3, q) = |-6 + 3q - 2q| = |q - 6|$, which vanishes never and equals 1 exactly for $q = 5, 7$: precisely the members whose double branched cover is an integral homology sphere. This threshold reappears on the symplectic side.

3. THE LAGRANGIANS OF THE FAMILY

3.1. The pillowcase and the two curves. We use $\text{SU}(2)$ as the unit quaternions; a traceless element is a purely imaginary unit. The traceless character variety of the four-punctured sphere is the pillowcase of the introduction, with coordinates (γ, θ) , involution $\iota(\gamma, \theta) = (-\gamma, -\theta)$, and corners $\{0, \pi\}^2$ [11, 4]. We work on the double cover T^2 , where curves lift to ι -invariant families; the count bookkeeping is §4.2. A rational tangle Q_r has for its character variety a straight line of slope r through the corners [4]. Two operations build the family's curves:

- *Conway sum.* On the pillowcase, $T_1 + T_2$ is the fiber product over γ : the θ -coordinates add, and over each seam $\gamma \in \{0, \pi\}$ a circle of reducibles appears when both summands are non-abelian there [13, Thm. 4.22]. Perturbing resolves each circle by a diagonal cut-and-paste [13, Thm. 4.23], adding one self-intersection per circle.
- *Earring.* $\widehat{Q}_r$ replaces the arc of Q_r by the Herald–Kirk figure-eight: the doubled arc closing through the corner folds, with one pinch [6, Def. 6.6]; its bounding cochain vanishes [13, Lem. 5.6].

For the decomposition (3) the -2 band is constant, so $\mathbf{red} = \mathcal{R}^\natural(\widehat{Q_{-1/2}})$ is *independent of q* (one immersed circle, one pinch), while $\mathbf{blue} = \mathcal{R}_t(Q_{1/3} + Q_{1/q})$ varies: two seams, four resolved circles, a single immersed circle whose self-intersection count grows with q (forty at $q = 5$, eighty-two at $q = 7$, two hundred twenty-five at $q = 11$). We require $\gcd(q, 3) = 1$ (else corner circles appear); this excludes $q = 9, 15, \dots$ from the pillowcase analysis but not from Theorem 1.1.

3.2. Validation gates and the naive ranks. The reconstruction is checked before it is used. At $q = 5$, red and blue meet in nine points (one interior, two clusters of four near $\gamma = \pi$), the winding-number search (§4) finds exactly two immersed bigons with distinct vertices, and

$$\text{rank } HF((\mathcal{R}^\natural, 0), (\mathcal{R}_t, 0)) = 9 - 2 \cdot 2 = 5,$$

matching [13, Thm. 5.9] in every particular. For the other members Smith gives no numbers; the gates are internal: generator and bigon counts are recomputed at two independent perturbations, and only perturbation-stable ranks are reported. (Generator counts themselves are not invariants and do vary — e.g. 19 vs 23 at $q = 11$, 25 vs 27 at $q = 13$ — but the homology ranks agree in every tested case. One non-generic earring perturbation at $q = 7$, which collapses the thirteen generators to nine, was identified and excluded.) The measured naive ranks are those of Table 1: 5, 5, 7, 15, 17 for $q = 3, 5, 7, 11, 13$, giving Computation 1.2 against Theorem 1.1. The $q = 3$ value is the Hedden–Herald–Kirk computation of 8_{19} [5, §11.6], where the naive theory already equals $I^\natural$ and the cochain is trivial.

4. THE COMBINATORIAL FLOER PACKAGE

4.1. Polygons by winding numbers. For immersed curves in a surface, Floer differentials count immersed bigons, detected combinatorially by winding functions [3]. Fix generators x, y . A candidate bigon from x to y is a loop $\Gamma = \alpha \cdot \beta$ with $\alpha \subset \mathbf{red}$ from x to y and $\beta \subset \mathbf{blue}$ from y to x ; lift to T^2 and let w be the winding function. Then Γ bounds an immersed bigon iff (a) $[\Gamma] = 0 \in H_1(T^2)$; (b) $w \equiv 0$ at the four pillowcase corners; (c) $w \geq 0$ on every complementary region; (d) the corners at x, y are convex (single embedded sector of angle $< \pi$ with winding jump $+1$). The differential is $\partial x = \sum_y (\#\{\text{bigons}\} \bmod 2) y$ and $\text{rank } HF = \#\{\text{generators}\} - 2 \text{rank } \partial$ over $\mathbb{F}_2$. The same conditions, with a convex-corner test at *every* vertex, detect the higher immersed polygons below. Delicate numerical points (long edges under mod- 2π arithmetic, shallow corners, adaptive sampling) are handled as recorded in the source.

4.2. Lift bookkeeping. A pillowcase polygon lifts to exactly two ι -related T^2 polygons; fixing the lift C_+ of red selects exactly one, so pillowcase counts equal T^2 counts with red arc on C_+ , with no division. A pillowcase self-intersection of blue has two T^2 preimages, and a polygon may close through either; contributions are accumulated over both. One pitfall is flagged for reproducibility: some generators fold under $T^2 \rightarrow P$, so proximity in P is not proximity on the cover; all locality pruning must be done in blue-arc position on the cover. (In an intermediate version of this work a pillowcase-metric pruning silently dropped a genuine quadrilateral; the corrected runs are what is reported here.)

4.3. The deformed differential and the Maurer–Cartan equation. For $b = \sum_i c_i s_i$ ($c_i \in \mathbb{F}_2$) on blue, the deformed differential counts immersed polygons with two generator vertices and the remaining vertices at crossings in the support [2, §2.9], [13, §5]:

$$\partial_b(x) = \sum_y \left(\#\mathcal{B}(x, y) + \sum_i c_i \#\mathcal{T}(x, s_i, y) + \sum_{i < j} c_i c_j \#\mathcal{Q}(x, s_i, s_j, y) + \dots \right) y \pmod{2}, \quad (6)$$

with $\mathcal{B}, \mathcal{T}, \mathcal{Q}$ the bigons, triangles, quadrilaterals (μ^1, μ^2, μ^3). The Maurer–Cartan equation (2) is the analogous count with all vertices at self-crossings: monogons μ^0 , self-bigons μ^1 , self-triangles

μ^2 , etc. The polygon counter was validated on Smith's unlink prototype (where a single triangle through the earring pinch is the known mechanism).

Proposition 4.1 (Unobstructedness). *The blue Lagrangians bound no teardrops: $\mu^0 = 0$, so (2) is a genuine equation.*

Proof. A teardrop is a Maslov-index-one disk with one corner, placing μ^0 in degree 2 of $CF(\mathcal{R}, \mathcal{R})$; but a transverse self-intersection of a curve in a surface contributes generators in degrees 0 and 1 only [1], so $CF^2 = 0$. (The argument requires disks in the smooth stratum, where a $\mathbb{Z}$ -grading exists; computationally every self-intersection lies at distance ≥ 0.6 from every corner. Direct monogon counts confirm $\mu^0 = 0$, and the counter detects a genuine teardrop on a test curve.) This is the standard mechanism by which graded curves in surfaces are unobstructed; we record it because it is what makes the deformed theory well posed. $\square$

The higher self-products vary across the family and matter: at $q = 5$ also $\mu^1 = 0$ and $\mu^2 = 0$ (all pairs and all $\binom{40}{3}$ window-pruned triples tested), so Maurer–Cartan is vacuous on small support; at $q = 7$ and $q = 11$ the self-differential μ^1 is nonzero (two self-bigons each), so (2) genuinely constrains b , and the computed cochains below pass it.

5. THE THREE COCHAINS

5.1. $q = 5$: unique cancellation through a quadrilateral. The naive differential has rank 2 (two bigons, pairs (x_1, y_0) and (x_4, y_6)); the target rank is 1. Over $\mathbb{F}_2$ a deformation reaches rank 1 only by cancelling one bigon entry exactly (the added polygon terms must land on it). No triangle through any of the forty crossings reaches either bigon pair; the cancellation happens one order up. There is an immersed quadrilateral with boundary vertices x_4, y_6 and interior vertices s_A, s_B (window-stable, hence a genuine immersed disk); neither crossing carries any other polygon, so for $b = s_A + s_B$, $\partial_b = \partial_0 + E_{x_4, y_6}$ has the single entry (x_1, y_0) and rank $HF = 9 - 2 = 7 = \text{rank } I^\natural$. A search over all supports by increasing size shows no single crossing and no other pair achieves rank 1 — e.g. the geometrically nearest cancelling pair drags in three triangles and lands at rank 3 — so $b = s_A + s_B$ is the unique minimal solution; by the vacuity of Maurer–Cartan on small support it is a bounding cochain. This proves Computation 1.3(i).

5.2. $q = 7$: unique cancellation through a triangle. Thirteen generators, three bigons, naive rank 7; target $9 = \text{rank } I^\natural$ means rank $\partial_b = 2$: again a cancellation. Here a triangle suffices: a single self-crossing $s \approx (0.05, 5.41)$ carries an immersed triangle whose generator pair coincides with one bigon of ∂_0 ; s carries no other generator-touching polygon and is μ^1 -closed (it is the tail of no self-bigon), so $b = s$ satisfies (2) and cancels that bigon: rank $HF = 13 - 4 = 9$. The search over all single crossings, filtered by the Maurer–Cartan check, returns exactly this one; it is the unique minimal solution, and it recurs at the same pillowcase coordinate under change of perturbation. This proves Computation 1.3(ii). The contrast with $q = 5$ — triangle where a quadrilateral was forced — shows the polygon order of the correction is not a family invariant even within the deficit regime.

5.3. $q = 11$: abundant creation. At the reported perturbation the complex has nineteen generators and naive rank $15 > 13 = \text{rank } I^\natural$: the correction must *create* one differential, i.e. raise rank ∂_b from 2 to 3. Creation is combinatorially generic: any immersed triangle through a single crossing whose generator pair is *not* already a differential adds a new entry. Enumerating all triangles through the 225 crossings (108 of them carry triangles), evaluating rank ∂_b , and filtering by the Maurer–Cartan check ($\mu^0 = 0$ everywhere; $\mu^1 \neq 0$, so μ^1 -closedness is a real constraint) yields *fifty-five* single-crossing bounding cochains with deformed rank 3, i.e.

$\text{rank } HF = 19 - 6 = 13 = \text{rank } I^{\natural}$. Notably, in fifty of the fifty-five the created differential flows into the same distinguished generator. This proves Computation 1.3(iii).

Remark 5.1 (Rigidity asymmetry). At deficit members the rank condition pins the cochain uniquely; at the excess member it manifestly does not. This is forced by the geometry of (6): a cancelling polygon must land on an occupied entry (a strong incidence condition on its vertices), while a creating polygon may land on any empty entry. Consequently, at excess members the bounding cochain of the *tangle* — the object the CHKK conjecture actually attaches — is not determined by rank-matching with $I^{\natural}$; any further pinning must come from the naturality/invariance data of the conjecture itself. We regard this as a genuine, and previously invisible, feature of the correction theory.

6. DISCUSSION

6.1. The law and the threshold. Computation 1.2 says the naive pillowcase theory is never more than one differential from the truth on this family, with the error direction governed by $\det \geq 3$. Two reformulations suggest where a proof might come from. First, $\det = |q - 6| < 3$ happens exactly when the double branched cover $\Sigma_2(K_q)$ is an integral homology sphere; the deficit regime is the homology-sphere regime. Second, with $d = (\det - 1)/2$ the number of binary-dihedral traceless characters, the law reads naive $- I^{\natural} = 2 \operatorname{sgn}(d - 1)$: one binary-dihedral character is exactly compensated; zero leaves a deficit; two or more leave an excess. Since the binary-dihedral characters sit at the pillowcase seams — where the Conway-sum fiber circles are resolved — a curve-level explanation seems within reach of the combinatorial model, and we pose it as the main open problem: *prove the deficiency law for all odd q coprime to 3*. A proof would convert Table 1 from data into the first family-wide statement about the size of the bounding-cochain correction.

6.2. Relation to prior examples. The one previously exhibited bounding cochain — CHKK’s, on the earring tangle of $T(4, 5)$ [2, §10] — acts by creation (rank $9 \rightarrow 7$). Our deficit members show that Conway-sum cochains can act by cancellation, our excess member shows they can also act by creation, and the family shows both behaviors occur on the *same* tangle type, separated by the determinant. The mechanism is thus not tied to the earring versus Conway-sum dichotomy, as the two prior data points might have suggested, but to the deficiency sign.

6.3. Reproducibility. All computations are pure Python (standard library), included with the source; each module is a self-checking battery. The Alexander recursion is validated by the eleven-coefficient Lehmer match at $q = 7$; the curve reconstruction by Smith’s $q = 5$ numbers; the polygon counter by the unlink prototype and synthetic teardrop/lens tests; the naive ranks and cochains by independent perturbations. $\text{rank } I^{\natural}$ in Table 1 is Theorem 1.1 and is independent of all of this.

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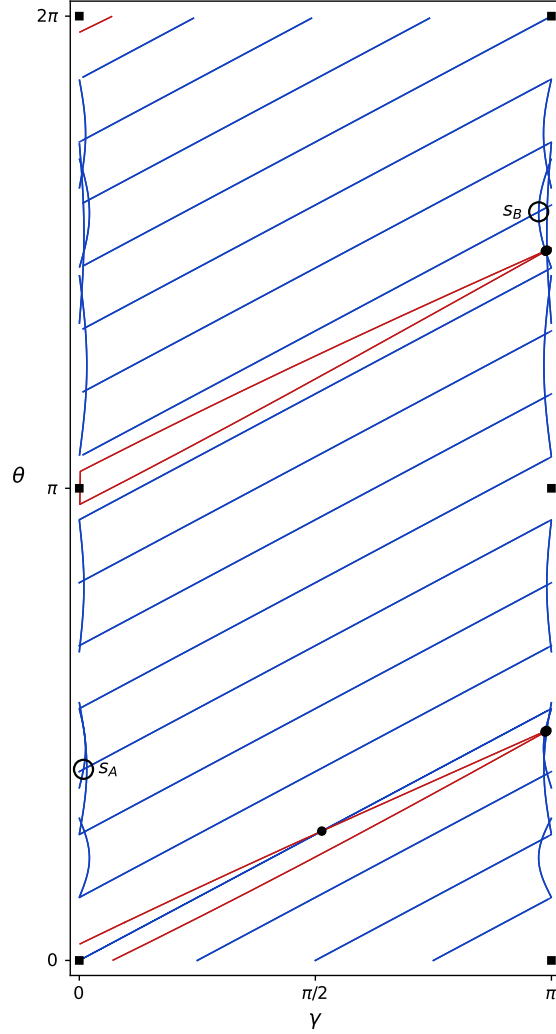


FIGURE 1. The two Lagrangians at $q = 5$ on the fundamental domain $[0, \pi] \times [0, 2\pi)$ of the pillowcase, plotted from the computed curves themselves (the pillowcase counterpart of Smith's Figure 31). Blue is the perturbed Conway sum $\mathcal{R}_t(Q_{1/3} + Q_{1/5})$, a single immersed circle with forty self-intersections clustered along the seams. The small sinusoidal arcs hugging the seams $\gamma \in \{0, \pi\}$ are the cut-and-paste connectors resolving the four seam fiber circles, drawn at the perturbation amplitude $\varepsilon \approx 0.05\text{--}0.07$; these arcs carry the self-crossings on which the bounding cochain lives, and the V-shaped turnbacks at the border are strands reflecting off the seams in the quotient. Red is the earring figure-eight $\mathcal{R}^{\natural}(\widehat{Q_{-1/2}})$, the doubled line of direction $(2, 1)$ closing through the corner folds. Across the interior the blue strands run *nearly parallel* to red (slopes $8/15$ vs $1/2$), so each interior band is a red line riding a blue one — this near-tangency (crossing angle ≈ 0.026) is why the interior generator exists and why shallow-angle corner tests are needed in the computation. Solid dots are the nine generators: one interior, and two clusters of four near $\gamma = \pi$ that merge at this scale. The two circled self-intersections of blue are the support s_A, s_B of the $q = 5$ bounding cochain of Computation 1.3(i), one on each seam, exchanged by $(\gamma, \theta) \mapsto (\pi - \gamma, -\theta)$. Curves appear as several arcs because the fundamental domain folds the double cover.